Proposal Appendices: Resources & Others

Supplementary Figures

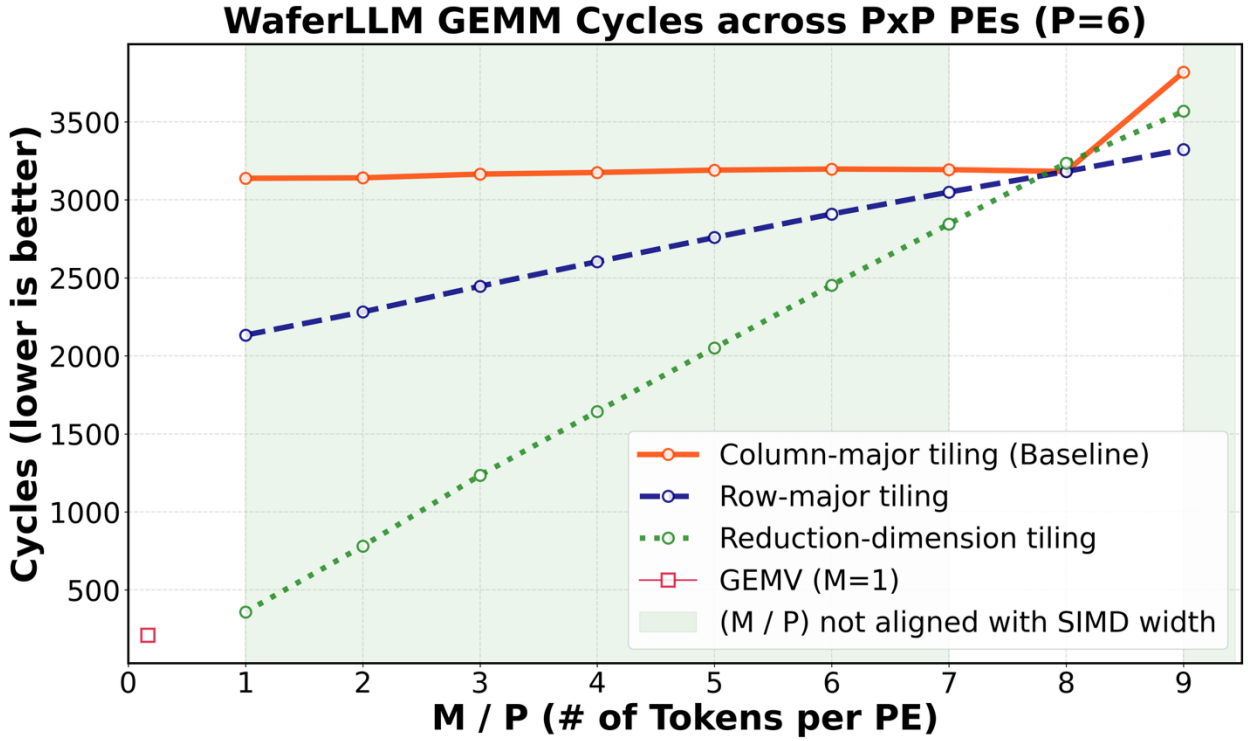

Figure 1. Latency comparison, measured in cycles, between column-major, row-major, and reduction-dimension tiling methods, as well as GEMV, across varying token counts.

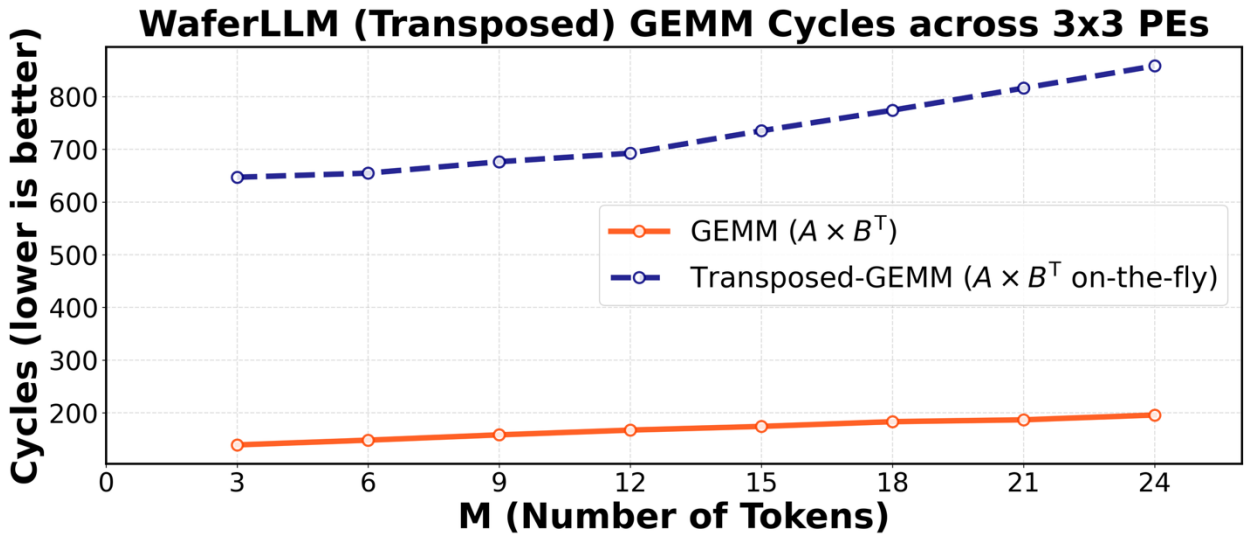

Figure 2. Latency comparison, measured in cycles, between the original GEMM and the Transposed-GEMM used in WaferLLM across varying token counts.

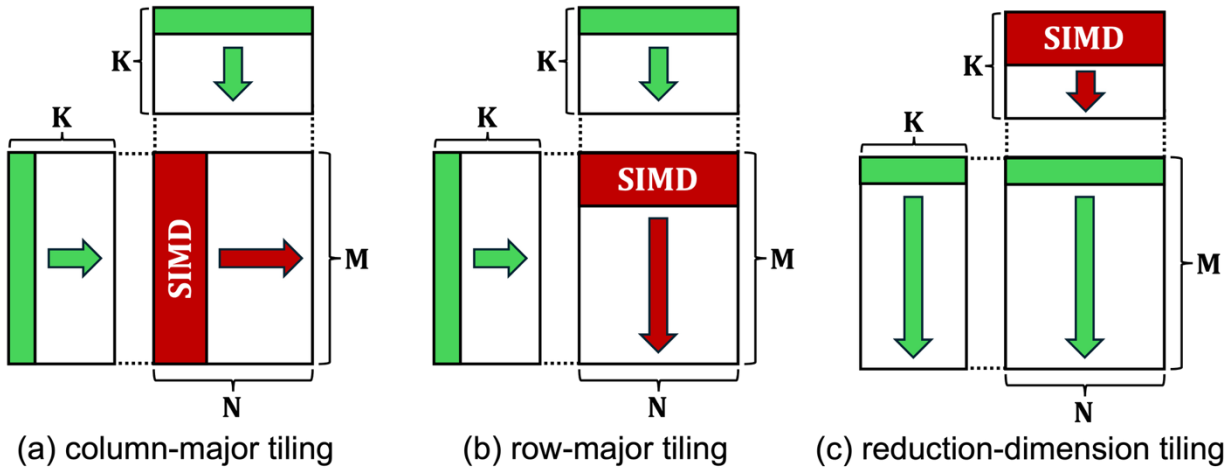

Figure 3. Illustration of three GEMM tiling strategies: (a) column-major, (b) row-major, and (c) reduction-dimension, used in the Cerebras SDK to compute $C_{MN} = A_{MK} \times B_{KN}$. Green arrows indicate the outer-loop traversal, while red arrows denote the inner-loop direction optimized with the @map directive, where each inner-loop executes in parallel using an 8-lane SIMD.

References

[1] T. Dao, D. Fu, S. Ermon, A. Rudra, and C. Ré, “FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness,” in Advances in Neural Information Processing Systems, vol. 35, 2022, pp. 16344-16359.

[2] A. Agrawal, N. Kedia, A. Panwar, J. Mohan, N. Kwatra, B. Gulavani, A. Tumanov, and R. Ramjee, “Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve,” in 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24), 2024, pp. 117-134.

[3] A. K. Kamath, R. Prabhu, J. Mohan, S. Peter, R. Ramjee, and A. Panwar, “POD-Attention: Unlocking Full Prefill-Decode Overlap for Faster LLM Inference,” in Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2, 2025, pp. 897-912.

[4] R. Zhu, Z. Jiang, C. Jin, P. Wu, C. A. Stuardo, D. Wang, X. Zhang, H. Zhou, H. Wei, Y. Cheng, J. Xiao, X. Zhang, L. Liu, H. Lin, L.-W. Chang, J. Ye, X. Yu, X. Liu, X. Jin, and X. Liu, “MegaScale-Infer: Efficient Mixture-of-Experts Model Serving with Disaggregated Expert Parallelism,” in Proceedings of the ACM SIGCOMM 2025 Conference, 2025, pp. 592-608.

[5] C. He, Y. Huang, P. Mu, Z. Miao, J. Xue, L. Ma, F. Yang, and L. Mai, “WaferLLM: Large Language Model Inference at Wafer Scale,” in 19th USENIX Symposium on Operating Systems Design and Implementation (OSDI 25), 2025, pp. 257-273.

[6] J. Li, Y. Jiang, Y. Zhu, C. Wang, and H. Xu, “Accelerating Distributed MoE Training and Inference with Lina,” in 2023 USENIX Annual Technical Conference (USENIX ATC 23), 2023, pp. 945-959.